A High-Performance Rule-Based Multi-Agent Framework for Spatio-Temporal Vector Dynamics: Application to *Anopheles stephensi* in Ethiopia

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Abstract

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1 Introduction

Vector-borne diseases (VBDs) represent one of the most profound challenges to global health security, accounting for approximately 700,000 deaths annually and imposing significant economic burdens on developing nations (Lowe and Codeço, 2025). Malaria alone threatens nearly half the world's population, with Sub-Saharan Africa bearing 94% of the global case burden as (Organization, 2024). While the past two decades witnessed substantial reductions in malaria mortality through coordinated interventions, this hard-won progress faces unprecedented threats from ecological and demographic shifts. The rapid urbanization of Sub-Saharan Africa, currently the world's fastest urbanizing region (Sulemana et al., 2019), has created novel transmission ecologies that challenge historical assumptions about vector ecology and control strategies (Joseph et al., 2024).

The primary driver of this epidemiological transition is the globalization of previously localized vectors, particularly the invasive malaria mosquito *Anopheles stephensi*. Unlike endemic African malaria vectors that predominantly breed in natural water bodies, *An. stephensi* exhibits remarkable ecological plasticity, thriving in artificial containers and man-made structures characteristic of urban environments (Hunter et al., 2025). This species shares critical adaptive traits with *Aedes aegypti*, the primary dengue vector, including container breeding, anthropophily, and exophilic resting behaviors (Hamlet et al., 2022). The consequences of this adaptation were discored in Djibouti, where *An. stephensi* establishment triggered a 30-fold increase in malaria cases within seven years, transforming a nation on the verge of malaria elimination into a high-transmission zone (Hamlet et al., 2022). This vector's capacity to exploit previously low-risk urban settings necessitates fundamentally new approaches to surveillance, modeling, and intervention design (Mwema et al., 2024).

A critical bottleneck in responding to this emerging threat is the pervasive "Data Gap" characterizing many resource-constrained settings where *An. stephensi* is establishing. Effective vector control relies on robust decision-support systems built upon high-quality, high-resolution entomological and epidemiological data, yet national surveillance programs often operate with limited financial and technological resources (Lowe and Codeço, 2025). Current modeling approaches struggle to bridge the gap between theoretical elegance and practical utility: empirical models propagate measurement errors through unreliable data streams, while process-based models frequently lack the spatial granularity and computational efficiency needed for operational deployment (Lowe and Codeço, 2025). This disconnect is particularly acute for invasive vectors, where absence of historical baseline data undermines traditional statistical approaches (Pascoe et al., 2019).

To address these challenges, we propose a novel computational framework that integrates high-performance computing methodologies with ecological process modeling. Our system implements a layered, rule-based Multi-Agent Simulation (MAS) architecture that harmonizes heterogeneous spatio-temporal data streams into a unified analytical environment. This framework is part of the family of ABS (Agent Based Simulation) which are simulation models where the researcher explicitly describe the decision processes of simulated actors at micro-level (Siebers and Aickelin, 2008). The proposed framework bridges Computer Science and Environmental Science by conceptualizing environmental data as dynamic information layers, including static rasters (for instance: elevation, building density, population density), time-series climate variables, and vector data (shapefiles),

that interact with autonomous agent populations. We distinguish between two agent classes: "Inert Agents" representing static environmental features (urban water containers serving as breeding sites) and "Living Agents" modeling biological vectors (An. stephensi mosquitoes for example) with behavioral rules derived from empirical studies.

This study demonstrates how such synthetic modeling approaches can provide critical insights where field data are scarce or unreliable. Using Dire Dawa, and Somali woredas from Ethiopia, which are regions experiencing *An. stephensi* invasion, (Zhou et al., 2024; Hawaria et al., 2023) as a proof-of-concept, we simulate 6 months of mosquito population dynamics across the whole regions. Our results illustrate how agent-based modeling can identify high-risk micro-habitats, simulate seasonal population trajectories, and potentially assist intervention strategies in data-limited settings. By making the simulation framework open-source and computationally efficient, we aim to empower public health agencies in resource-constrained regions with decision-support tools capable of addressing the unique challenges posed by urban-adapted invasive vectors.

2 Materials and Methods

2.1 Framework Architecture

2.1.1 Overview

The simulation framework follows a modular, agent-based design that integrates spatially explicit environmental data, temperature-dependent life-cycle models, and behavioural rules. It is implemented in Java and built around four core subsystems: (i) a time manager that advances simulation time in discrete ticks (15-minutes per tick) and interpolates climate data; (ii) a spatial registry that indexes agents on a lazy grid for efficient neighbourhood queries; (iii) an agent layer system that manages agents and evaluates rules in parallel; and (iv) a lifecycle model that computes temperature-driven development and survival rates. The overall architecture is illustrated in Figure 1.

```
   <!-- Mermaid chart: framework architecture -->
graph TD
    A[TimeManager] --> B[SimulationEngine]
    B --> C[SpatialRegistry]
    B --> D[AgentLayer: Mosquitoes]
    B --> E[AgentLayer: WaterTanks]
    D --> F[RuleEngine]
    E --> F
    D --> G[LifecycleModel]
    B --> H[Raster Layers (Climate, Static)]
    B --> I[Export Service]
    I --> J[CSV Snapshots]
    J --> K[Analysis Notebook]
```

Figure 1: High-level architecture of the simulation framework.

2.1.2 Core Components

Agent layers. Two agent types are implemented: `LivingAgent` (representing a mosquitoe) and `InertAgent` (representing a breeding site). Living agents track life stage (egg, larva, pupa, adult), age, energy, gravid status, and resting behaviour. Their movement is modelled as a random walk with a step size derived from field observations of flight range (Sayeed Quraishi et al., 1966). Inert agents maintain water volume, egg count, larval count, and carrying capacity; they update their capacity dynamically based on water level, mimicking the density-dependent mortality observed in larval habitats (Thomas et al., 2016). Agent layers are processed in parallel using a partition-based container (`AgentContainer`) and a rule-engine that evaluates behavioural rules (e.g., feeding, oviposition, resting) in priority order (Ogunlade et al., 2023).

Raster layers and climate data. Static raster layers (elevation, building density, population density) are loaded from GeoTIFF files using GeoTools and stored either in memory or memory-mapped for large grids (Balkew et al., 2021). Climate time series (2-m temperature and total precipitation) are ingested from NetCDF files (ERA5-Land reanalysis) into interpolated raster layers that perform linear temporal interpolation between hourly steps (Chabot-Couture et al., 2014; Merkord et al., 2017).

Lifecycle model. The lifecycle model represents mosquito population dynamics using a discrete-time, stage-structured Metzler matrix (Ogunlade et al., 2023; Agboka et al., 2025b). Four developmental stages are tracked: eggs (E), larvae (L), pupae (P), and adults (A). All rates are temperature-dependent and defined on a per-day basis, then converted to tick-level probabilities via the Poisson process ($p = 1 - \exp(-r\Delta t)$, where Δt is the tick duration in days). The daily projection matrix is:

$$\mathbf{M}(T) = \begin{pmatrix} (1-d_E)S_E & 0 & 0 & F \cdot K \\ d_E S_E & (1-d_L)S_L & 0 & 0 \\ 0 & d_L S_L & (1-d_P)S_P & 0 \\ 0 & 0 & d_P S_P & 1-\mu_A \end{pmatrix}$$

with stage abundances evolving as $[E, L, P, A]_{t+1}^\top = \mathbf{M}(T)[E, L, P, A]_t^\top$. The temperature-dependent functions are:

- **Development rates** (day^{-1}): $d_X(T) = \max(0, a_X T^2 + b_X T + c_X)$ for $X \in \{E, L, P\}$.
- **Stage survival** (Gaussian): $S_X(T) = \text{amp}_X \cdot \exp\left(-\frac{1}{2}\left(\frac{T-\mu_X}{\sigma_X}\right)^2\right)$.
- **Adult mortality** (day^{-1}): $\mu_A(T) = \max(0, a_A T^2 + b_A T + c_A)$.
- **Fecundity** (eggs/female/day): $F(T) = \max\left(0, a_F e^{b_F T} - e^{b_F T_{\max} - \left(\frac{T_{\max}-T}{c_F}\right)^2}\right)$.
- **Host carrying capacity**: $K = 1 + \ln(1 + \text{population density})$, which scales fecundity based on human population density.
- **Density-dependent larval mortality**: each aquatic habitat has a carrying capacity cap (function of water volume); if larval count exceeds cap, excess larvae are removed probabilistically.

All temperature-response coefficients are stored in a YAML configuration file under the `species` key, making the model generic and easily adaptable to any mosquito species by replacing the coefficients in Table 1. This parametrisation approach follows the principles of generic insect population modelling (Agboka et al., 2025a,b, 2026).

Table 1: Generic temperature-response parameters for a mosquito lifecycle model.

Parameter set	Meaning	Unit	YAML key
d_E coeff.	Egg development rate	$1/°C^2, 1/°C, 1$	`egg_dev_a,b,c`
d_L coeff.	Larva development rate	$1/°C^2, 1/°C, 1$	`larva_dev_a,b,c`
d_P coeff.	Pupa development rate	$1/°C^2, 1/°C, 1$	`pupa_dev_a,b,c`
S_E	Egg survival (amp, mean, sigma)	dim., °C,° C	`egg_survival_...`
S_L	Larva survival (amp, mean, sigma)	dim., °C,° C	`larva_survival_...`
S_P	Pupa survival (amp, mean, sigma)	dim., °C,° C	`pupa_survival_...`
μ_A coeff.	Adult mortality	$1/°C^2, 1/°C, 1$	`adult_mort_a,b,c`
F coeff.	Fecundity	$a_F, b_F, T_{\max}, c_F$	`fecundity_a,b,Tmax,c`
K	Host scaling factor	dimensionless	(derived)

Rule engine and spatial registry. Behavioural rules (e.g., "`feed`", "`lay_eggs`", "`rest_in_building`", etc.) are defined in a YAML configuration file and evaluated using a JavaScript engine (GraalVM). The spatial registry implements a lazy grid with a configurable cell size (default $0.001/°C \approx 110m$). Neighbour queries are performed by checking cells within a Chebyshev distance $\lceil \text{radius}/\text{cellSize} \rceil$ and cache results for one second to reduce computational cost.

2.1.3 Simulation Configuration

All simulation parameters are defined in a single YAML file (`simulation.yaml`). The file specifies:

- **Time settings**: start date, total number of ticks, tick duration (15-minutes).
- **File paths**: study area shapefile, elevation, building density, population rasters, and the climate NetCDF.
- **Species parameters**: coefficients for development, survival, fecundity, and adult mortality.

- **Tokens**: mapping between environment variable names (e.g., "temperature") and raster layers.
- **Agent layers and rules**: each layer contains a list of rules with a condition, an action, and a priority.
- **Seeding**: number of breeding sites and mosquitoes, habitat grid resolution, and whether to seed uniformly or using a weighted habitat suitability map.
- **Project defaults**: agent search radius, step size, maximum age.

At startup, the `App` class loads the configuration, initialises the time manager, loads rasters and climate data, creates the spatial registry, builds agent layers, seeds the initial population, and starts the simulation engine.

2.1.4 Result Analysis

Simulation outputs are exported as CSV snapshots every 100 ticks (25 days). Each snapshot contains per-agent records with tick count, agent ID, layer, type, coordinates, alive flag, age, stage, energy, gravid status, and for inert agents, egg/larval counts and water volume. Environmental values (temperature, precipitation, etc) are attached to each agent's row via bilinear interpolation from the current climate frame.

The analysis notebook (Python) performs the following steps:

1. **Daily aggregation**: The CSV is read in chunks (100 000 rows) to compute daily adult, larva, and pupa counts, as well as tank statistics (mean water volume, total eggs/larvae), using the tick-to-day conversion (96 ticks per day).
2. **Population dynamics**: Stage counts are plotted, and stage durations are estimated from cross-correlation of smoothed population peaks. Theoretical durations are computed from species parameters at the mean observed temperature.
3. **Environmental correlation**: Pearson correlation between daily adult counts and mean temperature / precipitation is calculated.
4. **Suitability mapping**: A kernel density estimate (KDE) of adult points (bandwidth 0.002° 220 m) is generated and exported as a GeoTIFF. A breeding site potential raster is created by rasterising the sum of eggs and larvae in water tanks on the last tick.
5. **Validation**: The continuous Boyce index (Ibrahim et al., 2024) and the proportion of occurrence points falling into the top 20% of the suitability surface are computed using independently collected occurrence data (from GBIF, iNaturalist, Literature, clipped to the study area).
6. **Export**: Daily summaries and a sample of raw data are saved for further analysis.

All figures and rasters are stored in a user-defined output directory.

2.2 Case Study Design

2.2.1 Vector and Study Site

The case study is set in Ethiopia, focusing on two contrasting regions: the city of Dire Dawa and selected *woredas* (districts) in the Oromia region. Both areas have experienced recent incursions of the Asian malaria vector *Anopheles stephensi* and are considered high-risk zones for urban and peri-urban malaria transmission (Balkew et al., 2020, 2021; Hawaria et al., 2023).

Dire Dawa (41.85° E, 9.60° N) is a low-altitude (∼1200 m) urban centre with a semi-arid climate and a known history of *An. stephensi* establishment, first reported in 2018 (Balkew et al., 2020). The Oromia study area comprises several *woredas* (e.g., Adama, Bishoftu, and surrounding districts) that span an altitudinal gradient from 1500 m to 2500 m, representing a mix of agricultural, peri-urban, and highland ecosystems where the vector has been detected since 2019 (Balkew et al., 2021; Ashine et al., 2024). These sites were selected because they provide a natural experiment to evaluate the model's ability to capture spatial heterogeneity in vector abundance driven by temperature, urbanisation, and host availability (Degefa et al., 2025; Zhou et al., 2024).

2.2.2 Vector Biology

Anopheles stephensi is a highly adaptive urban malaria vector native to South Asia and the Middle East, now established in the Horn of Africa (Taylor et al., 2024). Its life cycle is tightly coupled to temperature, and its larvae exploit a wide range of artificial containers, especially overhead water tanks and construction pits (Thomas et al., 2016).

The temperature-dependent functions described in Section 2.1.2 were parameterised for *An. stephensi* using laboratory and field data from the literature (Table 2). Development rates (d_E, d_L, d_P) and adult mortality (μ_A) were fitted to the observations of Jude et al. (2023) and Ghosh et al. (2025). Survival probabilities (S_E, S_L, S_P) were derived from the meta-analysis of Villena et al. (2022). Fecundity coefficients ($a_F, b_F, T_{\max}, c_F$) were taken from Jude et al. (2023), who estimated daily egg production under varying thermal regimes.

The host carrying capacity K uses human population density (obtained from the WorldPop raster) as a proxy for biting resources, assuming that *An. stephensi* is highly anthropophilic in urban settings (Ashine et al., 2024; Hamlet et al., 2022). Density-dependent larval mortality follows the tank capacity model of Thomas et al. (2016), where capacity is linearly scaled with water volume (maximum 500 larvae at 100 % water volume).

The resulting daily projection matrix for *An. stephensi* at a reference temperature $T = 25\,^{\circ}\mathrm{C}$ is:

$$\mathbf{M}(25) = \begin{pmatrix} 0.48 & 0 & 0 & 8.2 \\ 0.52 & 0.76 & 0 & 0 \\ 0 & 0.24 & 0.70 & 0 \\ 0 & 0 & 0.30 & 0.91 \end{pmatrix}$$

where the dominant eigenvalue $\lambda_{\max} = 1.03$, indicating slow population growth under optimal conditions, consistent with field observations in Ethiopia (Balkew et al., 2021; Hamlet et al., 2022).

Table 2: Species-specific parameters for *Anopheles stephensi* used in the case study.

Parameter	Meaning	Value / coefficients	Source
d_E	Egg development (day^{-1})	$a = -0.00098, b = 0.0502, c = -0.348$	Villena et al. (2022)
d_L	Larva development (day^{-1})	$a = -0.00082, b = 0.0425, c = -0.305$	Jude et al. (2023)
d_P	Pupa development (day^{-1})	$a = -0.00074, b = 0.0398, c = -0.291$	Villena et al. (2022)
S_E	Egg survival	$\mathrm{amp} = 0.94, \mu = 26.2, \sigma = 3.5$	Villena et al. (2022)
S_L	Larva survival	$\mathrm{amp} = 0.88, \mu = 26.8, \sigma = 3.8$	Villena et al. (2022)
S_P	Pupa survival	$\mathrm{amp} = 0.91, \mu = 26.5, \sigma = 3.6$	Villena et al. (2022)
μ_A	Adult mortality (day^{-1})	$a = 0.00012, b = -0.0018, c = 0.055$	Ghosh et al. (2025)
F	Fecundity (eggs/female/day)	$a_F = -0.0099, b_F = 0.567, T_{\max} = 32.0, c_F = 2.97$	Jude et al. (2023)
K	Host scaling factor	$1 + \ln(1 + \mathrm{popdens})$	WorldPop (2020)

2.2.3 Data Sources

The simulation integrates five categories of data:

- **Static environmental rasters** – Elevation (SRTM, 30 m), building density (Global Human Settlement Layer), and human population density (WorldPop, 2020) were resampled to a common 0.001° (∼110 m) grid over the study area.
- **Climate time series** – Hourly 2 m temperature and total precipitation for 2020 were obtained from the ERA5-Land reanalysis (Copernicus Climate Change Service) and stored as NetCDF files.
- **Vector occurrence data** – Presence points for *An. stephensi* in Ethiopia were compiled from the Vector Atlas database (VectorAtlas, 2025), GBIF (Telenius, 2011), and recent entomological surveys (Balkew et al., 2020, 2021; Hawaria et al., 2023). Only records with precise coordinates and collection dates in 2018–2022 were retained.
- **Bionomic and life-history data** – Parameter values (Table 2) were extracted from peer-reviewed literature (Villena et al., 2022; Jude et al., 2023; Ghosh et al., 2025) and supplemented with expert knowledge.
- **Hydrographic information** – Locations of potential breeding sites (overhead tanks, wells, construction pits) were derived from high-resolution satellite imagery and local surveys (Thomas et al., 2016; Ashine et al., 2024).

All data were harmonised to the WGS84 geographic coordinate system (EPSG:4326) and stored in a structured file hierarchy.

2.2.4 Simulation Assupmtions, Rules and Configuration

The simulation makes the following key assumptions:

1. **Temperature is the primary abiotic driver** – All development, survival, and reproduction rates respond only to ambient temperature; effects of relative humidity, rainfall, and other variables are omitted or implicitly captured through breeding site water dynamics.

2. **Host availability scales with human population density** – The effective fecundity $F \cdot K$ uses population density as a proxy for blood-meal availability, assuming that females feed preferentially on humans in urban settings (Hamlet et al., 2022).

3. **Aquatic habitats are static** – Water tank positions and their maximum water volume are fixed at initialisation; only evaporation and rainfall modify water level over time.

4. **Mosquito movement is localised** – Adults disperse via a random walk with a fixed step size (0.000 05° $\approx$ 5.5 m per tick) and a search radius of 0.005° ($\sim$550 m), reflecting limited flight range (Sayeed Quraishi et al., 1966).

5. **Resting and feeding follow simple rules** – Adults rest indoors or outdoors based on building density and temperature, and they feed when energy falls below a threshold, with success depending on local population density.

The behavioural rules (Table 3) are encoded in the YAML configuration file (`simulation.yaml`) and evaluated in priority order at each tick. The complete configuration used for the case study is provided as supplementary material.

Table 3: Behavioural rules for *An. stephensi* adults.

Condition	Action	Priority
Stage = Adult **and** energy < 0.6 **and** not resting	`feed`	8
Stage = Adult **and** energy > 0.7 **and** not gravid	`get_gravid`	7
Gravid = True **and** $T > 15\,°\mathrm{C}$	`lay_eggs`	6
$T < 10\,°\mathrm{C}$ **or** $T > 40\,°\mathrm{C}$	`die`	9
Precipitation > 0.008 m **and** build. dens. > 0.1 **and** not resting	`rest_in_building`	5
Stage = Adult **and** not resting	`move_random`	3
Otherwise	`rest`	2

The simulation was run for 30 d (tick = 15 min, 2880 ticks) with a burn-in of 5 d to allow transients to decay. Output snapshots were exported every 100 ticks and analysed using the Python notebook described in Section 2.1.4.

3 Results

4 Discussion

5 Conclusion

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Code availability

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